Lab Ten – Point Estimation

Tips

- Complete the tasks below. Make sure to start your solutions in on a new line that starts with "**Solution**:".
- Make sure to use the Quarto Cheatsheet. This will make completing and writing up the lab *much* easier.
- Make sure to use "enter" to make your code readable, so it doesn't run off the page. I switched the Quarto document to automatically render to .pdf, so you can see what you're submitting by default. You can switch back for the highlighting by moving the html block above the pdf block in the YAML header.
- Don't forget that you can copy-and-paste code when you're doing something similar as we did in class!
- Make sure to plot all computed probabilities. This is good `ggplot` practice AND will help make sure you don't make a mistake when computing the probability.

1 Lab 8 Notes

1.1 Altham's Multiplicative Binomial Distribution

The PMF for Altham's Multiplicative Binomial Distribution is

$$f_X(x) = \frac{1}{c}\binom{n}{x}p^x(1-p)^{n-x}\theta^{x(n-x)}I(x \in \{0,1,...,n\})$$

where

$$c = \sum_{i=0}^{n}\binom{n}{i}p^i(1-p)^{n-i}\theta^{i(n-i)}.$$

Just so we start on the same page, I have included my `R` function for computing probability using this distribution.

```r
daltbinom <- function(x, size, prob, theta) {
  normalizing.constant <- 0
  for (j in 0:size) {
    normalizing.constant <- normalizing.constant +
      choose(size, j) * prob^j * (1 - prob)^(size - j) * theta^(j * (size - j))
  }
  # Probability Mass at x=x
  (1 / normalizing.constant) *
    choose(size, x) *
    prob^x *
    (1 - prob)^(size - x) *
    theta^(x * (size - x)) *
    (x >= 0) *
    (x <= size)
}
```

2 Altham's Multiplicative Beta Binomial Distribution

The PMF for Altham's Multiplicative Beta Binomial Distribution is

$$f_X(x) = \left[\int_0^1 \left(\frac{1}{c}\binom{n}{x}p^x(1-p)^{n-x}\theta^{x(n-x)}\right)\left(\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}p^{\alpha-1}(1-p)^{\beta-1}\right)dp\right]I(x \in \{0,1,...,n\})$$

Just so we start on the same page, I have included my `R` function for computing probability using this distribution.

```
daltbbinom.single <- function(x, size, alpha, beta, theta) {
  integrand <- function(p) {
    sapply(
      p, # integrate will pass in a vector, so we use sapply to apply the following
      function(p_val) {
        # f(x|p) * f(p|alpha, beta)
        daltbinom(x, size, p_val, theta) * dbeta(p_val, alpha, beta)
      }
    )
  }
  # Probability Mass at x=x
  # integrate(integrand, lower = 0, upper = 1)$value * (x>=0)*(x<=size)
  # I shrink the bounds below to avoid log(0)=-infinty
  # I use try() to catch when there is an error and avoid crashing
  res <- try(
    integrate(integrand, lower = 1e-7, upper = 1 - 1e-7)$value *
      (x >= 0) *
      (x <= size),
    silent = TRUE
  )
  # If there is an error, return a near-zero probability
  if (inherits(res, "try-error")) {
    return(0.1^6)
  }

  res
}
# Write a function that works on a vector
daltbbinom <- function(x, size, alpha, beta, theta) {
  sapply(
    X = x,
    FUN = daltbbinom.single,
    size = size,
    alpha = alpha,
    beta = beta,
    theta = theta
  )
}
```

3 Question 1

In Lab 8, we use Altham's Multiplicative Binomial Distribution. Your job was to interpret the impact of θ in that equation. In this lab, we will model the number of legs won in a best of eleven (first to six) match.

To help us think about the interpretation, something that was challenging before, let's try something a little more manageable.

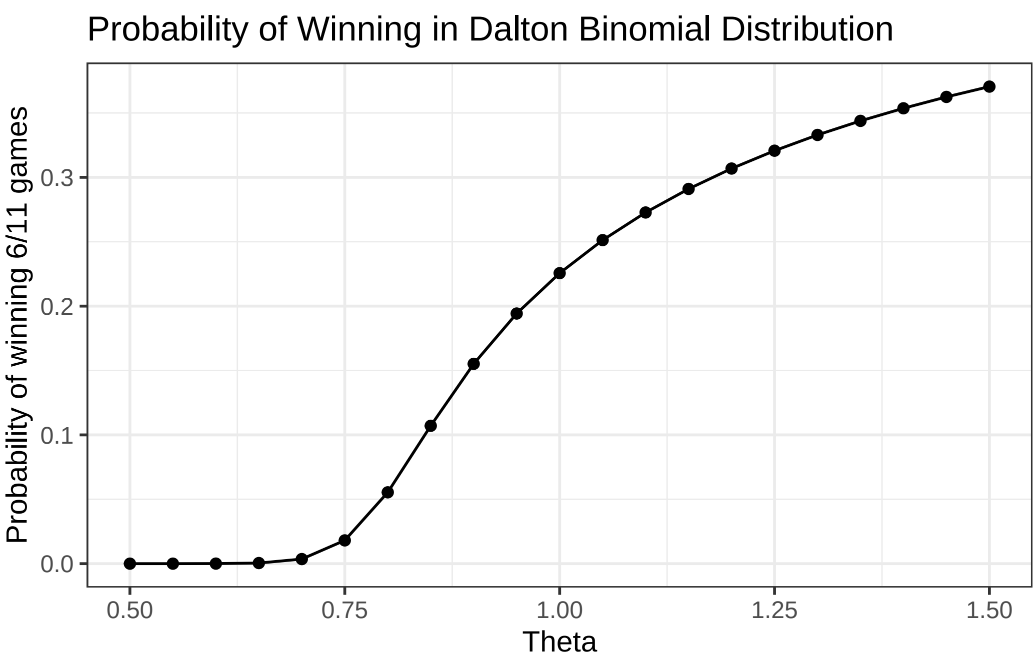
3.1 Part a

Suppose a player has a 50% chance of winning a specific leg, that is let $p = 0.50$. Let's look at the probability they win $(x = 6)$ legs in the match. Compute this probability over a sequence from $\theta = 0.5$ to $\theta = 1.5$ and plot it using a line where the x-axis is θ and the y-axis is $P(X = 6)$.

Solution

```r
# Create the vector of probabilities for each theta. We use a function to essentially "vectorize" the daltb

prob = function(t) {
  daltbinom(x = 6, size = 11, prob = 0.5, theta = t)
}


thetas = seq(from = 0.5, to = 1.5, by = 0.05 )

probs = sapply(thetas, prob)
data = tibble(thetas,probs)
```

```r
#Plot the probability vs theta

ggplot(data) + # Specify Data
geom_point(aes(x = thetas, # Specify x-axis
y = probs)) + # Specify y-axis
geom_line(aes(x = thetas, # Specify x-axis
y = probs), # Specify y-axis
method = "lm", # Specify best fit line
color = "black") + # Specify line color
theme_bw() + # A cleaner theme
labs(x = "Theta", # Specify axis labels
y = "Probability of winning 6/11 games") +
ggtitle("Probability of Winning in Dalton Binomial Distribution") # Plot title
```

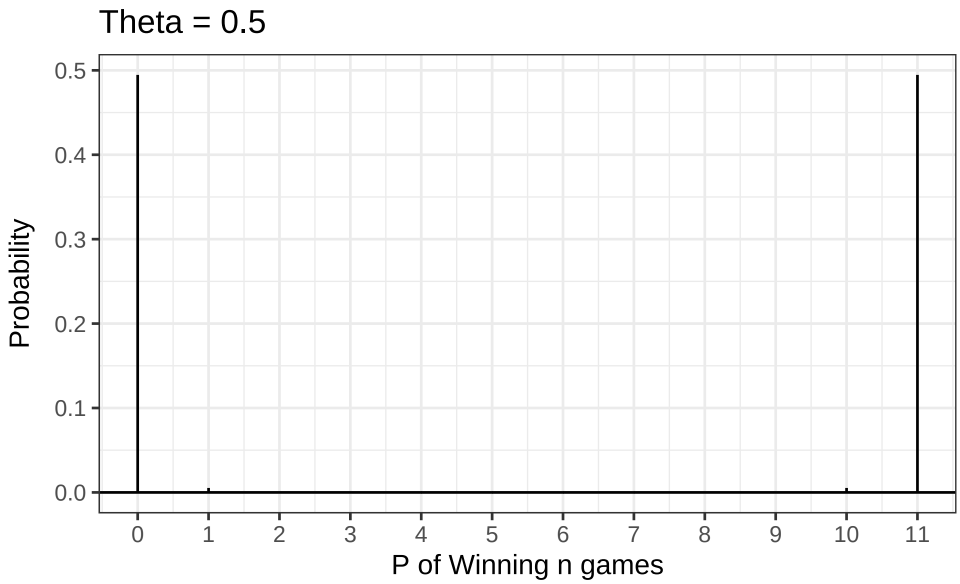


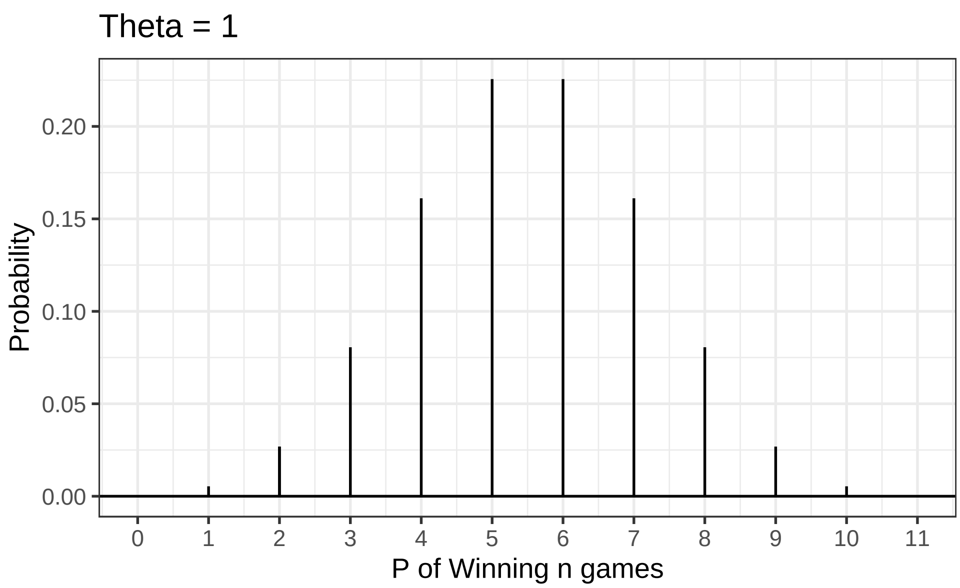
3.2 Part b

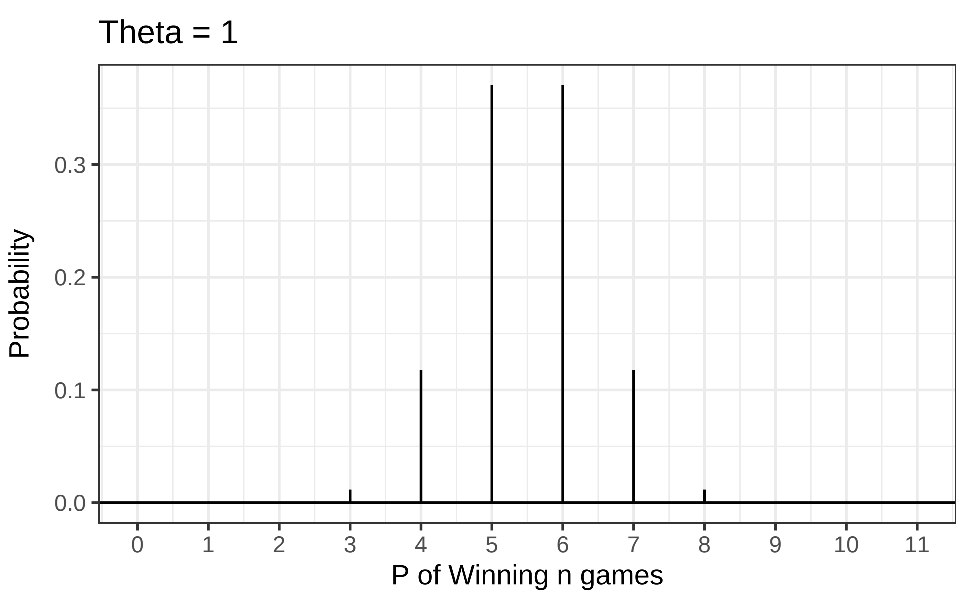
Now, plot Altham's Multiplicative Binomial Distribution under the same parameters with $\theta = 0.50$, $\theta = 1$, and $\theta = 1.5$. Use `ncol=1` in `facet_wrap()` to plot the PMFs in a single column for comparing.

```r
ggdat <- tibble(x = (0:11)) |>
  mutate(PMF05 = daltbinom(x = x, size = 11, prob = 0.5, theta = 0.5))

ggdat = ggdat |>
```

```r
      mutate(PMF01 = daltbinom(x = x, size = 11, prob = 0.5, theta = 1))

ggdat = ggdat |>
      mutate(PMF15 = daltbinom(x = x, size = 11, prob = 0.5, theta = 1.5))



PMF05.plot <- ggplot(ggdat) +
geom_linerange(aes(x = x, ymin = 0, ymax = PMF05)) +
geom_linerange(data = ggdat |> filter(x==2),
aes(x = x, ymin = 0, ymax = PMF05)) +
geom_hline(yintercept = 0) +
scale_x_continuous("P of Winning n games",
breaks = seq(0, 11, 1)) +
ylab("Probability") +
theme_bw()+
ggtitle("Theta = 0.5")


PMF01.plot <- ggplot(ggdat) +
geom_linerange(aes(x = x, ymin = 0, ymax = PMF01)) +
geom_linerange(data = ggdat |> filter(x==2),
aes(x = x, ymin = 0, ymax = PMF01)) +
geom_hline(yintercept = 0) +
scale_x_continuous("P of Winning n games",
breaks = seq(0, 11, 1)) +
ylab("Probability") +
theme_bw()+
ggtitle("Theta = 1")

PMF15.plot <- ggplot(ggdat) +
geom_linerange(aes(x = x, ymin = 0, ymax = PMF15)) +
geom_linerange(data = ggdat |> filter(x==2),
aes(x = x, ymin = 0, ymax = PMF15)) +
geom_hline(yintercept = 0) +
scale_x_continuous("P of Winning n games",
breaks = seq(0, 11, 1)) +
ylab("Probability") +
theme_bw()+
ggtitle("Theta = 1")

PMF05.plot / PMF01.plot / PMF15.plot
```


Theta = 0.5
Probability
0.0
0.1
0.2
0.3
0.4
0.5
0 1 2 3 4 5 6 7 8 9 10 11
P of Winning n games


Theta = 1
Probability
0.00
0.05
0.10
0.15
0.20
0 1 2 3 4 5 6 7 8 9 10 11
P of Winning n games


Theta = 1
Probability
0.0
0.1
0.2
0.3
0 1 2 3 4 5 6 7 8 9 10 11
P of Winning n games

3.3 Part c

What is your best assessment of what θ is doing in this setting. Consider when a 6-0 blowout is most likely, and when a 5-6 nail-biter is most likely. Note here we should treat the match as a fixed $n = 11$ as we did for $n = 3$ in Lab 8.
Note: In lab 8, we saw that small θ meant wins came in bunches (clumped together), so there were more 2-0 wins in a best of three series.

Solution

Here, we can see that theta is essentially a measure of how extreme the values will be in a distribution. A low theta (< 1) corresponds to an expectation of extreme outcomes, and a high theta (> 1) corrresponds to an expectation of moederate values. Here, this is interpreted as us expecting a close match, with both players winning a similar amount of games, when theta is low. Similarly, we expect blowouts when theta is high. When theta is moderate, we expect close-ish games, but accept that there will be some blowouts. I think that theta ≈ 1 is the most appropriate for this situation.

4 Question 2

4.1 Part a

Altham's Multiplicative Binomial Distribution does not typically simplify to the clean $E(X) = np$ seen in the Binomial distribution, unless $\theta = 1$. Write out the expression that would need to be solved to find $E(X^k)$, and then write a function that computes it given the proposed parameter values p (`prob`) and θ (`theta`).

Solution

We have the formula for the k^{th} moment:

$$E(X) = \sum_{x=-\infty}^{\infty} x f_x(x)$$

which we can compute here:

```r
Exk = function(k, size, prob, theta){
  x = 0:size
  fx = daltbinom(x = x, size = size, prob = prob, theta = theta)
  return(sum(x^k*fx))
}

Exk(1, 11, 0.5, 1)
```

```
[1] 5.5
```

```r
n = 11
p = 0.5

(n*p)
```

```
[1] 5.5
```

```r
# Our expected value of the altham binomial matches that of the binomial for theta = 1.
# This makes sense, as the Altham distribution for theta = 1 is the binomial distribution.
```

4.2 Part b

Describe how you can use your answer to part a to find Method of Moments estimates for the parameters p and θ for Altham's Multiplicative Binomial Distribution based on data.

Solution

From a dataset, we can obtain a first and second moment. We set the formula for the $1st$ moment (which is a function that takes arguments theta and p) equal to our observed value from the dataset. We do the same for the second moment. We now have two equations and two unknowns, and can use row reduction to solve for the two unknowns.

4.3 Part c

Write a function that computes the log-likelihood for Altham's Multiplicative Binomial Distribution, given data (`x`) and proposed parameter values p (`prob`) and θ (`theta`).

4.4 Part d

Describe how you can use your answer to part c to find Maximum Likelihood estimates for the parameters p and θ for Altham's Multiplicative Binomial Distribution based on data.

5 Lab 10

Peter "Snakebite" Wright is a two-time Professional Darts Corporation World Champion (2020, 2022) and a former World No. 1. He is one of the sport's most decorated players, winning nearly 50 PDC titles including the World Matchplay, UK Open, and multiple European Championships.

In 2024, he made The Premier League – a league involving the eight best players that runs every Thursday night from February to May. In this season, he won two of eighteen matches, securing only four points. The spread in legs won was -43, meaning he lost 43 more legs than he won. This performance was rather quite bad. He was soundly trounced in almost all his match ups.

The 2024 standings are below:

Rank	Player
1	Luke Littler
2	Luke Humphries
3	Michael van Gerwen
4	Michael Smith
5	Nathan Aspinall
6	Rob Cross
7	Gerwyn Price
8	Peter Wright

5.1 Summarize the 2024 Data

In `pdc2024.csv`, I have provided the data for the 2024 Premier League season. There are a few data cleaning steps to consider.

- Remove any matches with no `Score` (e.g., a bye or walkover)
- Nights 1 through 16 are best of 11 (first to 6), but the playoffs are extended. For consistency, keep nights 1 through 16 only.
- Make two columns `WinnerLegs` and `LoserLegs` that keep track of the number of legs each player has won

Summarize the data. What do the data tell us about the breakdown of the results?

5.2 Altham's Multiplicative Binomial Distribution

For each player in the 2024 season, use the number of legs won in each match to compute maximium likelihood estimates for p and θ.
Note: The "for each" here can be done more efficiently with a function or a loop!

- Create a vector containing the number of legs won by the player. Note sometimes you will need to pull from winner's legs and sometimes from the loser's legs.
- Find the MLE for the player using this data. Ensure to report p and θ for each player in a nicely formatted table.
  Note: In Lecture on Monday, we used transformations to restrict how `optim()` searches the parameter space. For example, here, you might consider `p <- 1/(1+exp(-par[1]))` and `theta <- exp(par[2])`.
- Make comments about the MLEs in the context of the standings.

5.3 Altham's Multiplicative Beta Binomial Distribution

For each player in the 2024 season, use the number of legs won in each match to compute maxmium likelihood estimates for p and θ.
Note: The "for each" here can be done more efficiently with a function or a loop!

- Create a vector containing the number of legs won by the player. Note sometimes you will need to pull from winner's legs and sometimes from the loser's legs.
- Find the MLEs for the player using this data. Ensure to report α, β, and θ for each player in a nicely formatted table.
  Note 1: Due to the computational complexity of `integrate()`, you should compute the logged probability for `0:6` once and add them according to the data in each player's vector. **Note 2**: In Lecture on Monday, we used transformations to restrict how `optim()` searches the parameter space.
- Plot the underlying Beta(α, β) distribution for each player.
  Note: Add the mean and variance of the Beta to the table of MLEs. Recall

$$E(X) = \frac{\alpha}{\alpha + \beta}$$

$$sd(X) = \sqrt{\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}}$$

- Make comments about the MLEs in the context of the standings.
  Note: The players alternate who throws first in a leg has the mathematical "head start" to reach a finish before their opponent can take another turn. Players alternate who throws first with each leg.

5.4 Executive Summary

Summarize the work you've done here to provide a brief (200-300 word) data-driven summary of the 2024 Premier League season using your work in Lab 10. Your summary should be professional, clear, and all claims should be corroborated with statistical support.